

4-Bit to 2-Bit Priority Encoder Using Dataflow Modeling

Aim

To design and verify a 4-bit-to-2-bit priority encoder in Verilog using dataflow modeling.

Code

```
`timescale 1ns/1ps

module priority_encoder_4to2 (
    input wire I3,
    input wire I2,
    input wire I1,
    input wire I0,
    output wire Y1,
    output wire Y0,
    output wire valid
);

assign Y1    = I3 | ((~I3) & I2);
assign Y0    = I3 | ((~I3) & (~I2) & I1);
assign valid = I3 | I2 | I1 | I0;

endmodule
```

Truth Table / Observation

Priority: $I3 > I2 > I1 > I0$

I3	I2	I1	I0	Selected Input	Y1	Y0	Valid
0	0	0	0	None	0	0	0
0	0	0	1	I0	0	0	1
0	0	1	0	I1	0	1	1
0	0	1	1	I1	0	1	1
0	1	0	0	I2	1	0	1
0	1	0	1	I2	1	0	1
0	1	1	0	I2	1	0	1
0	1	1	1	I2	1	0	1
1	0	0	0	I3	1	1	1
1	0	0	1	I3	1	1	1
1	0	1	0	I3	1	1	1
1	0	1	1	I3	1	1	1
1	1	0	0	I3	1	1	1
1	1	0	1	I3	1	1	1
1	1	1	0	I3	1	1	1
1	1	1	1	I3	1	1	1

Observation: All 16 input combinations were tested and matched the expected priority encoding.

Conclusion

The 4-bit-to-2-bit priority encoder was successfully implemented using Verilog dataflow modeling. The highest-priority active input is correctly encoded at the output, and the `valid` signal correctly indicates whether any input is active. All 16 input combinations passed simulation.